

Muskan Yadav

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Summary

Data Analyst | AI Enthusiast | Integrated M.Tech (CSE – Computational & Data Science), VIT Bhopal (CGPA: **7.98/10**). Skilled in **Python, SQL, Power BI, Machine Learning, Data Analytics, Data Visualization, Excel, and AI**. Built **3+** end-to-end AI & software projects, transforming data into actionable insights and developing scalable, data-driven solutions.

Education

Vellore Institute of Technology, Bhopal
Integrated M.Tech (B.Tech + M.Tech) in CSE

2022 – 2027
CGPA: **7.98/10**

Experience

Data Science Intern

Dec 2024 – Feb 2025

CodeSoft

- Developed machine learning pipelines using **Scikit-learn**; solved **5+** real-world business problems through iterative model tuning.
- Performed data analysis and manipulation using **Pandas** and **NumPy** across multiple structured datasets to surface key trends.
- Executed **ETL** workflows — preprocessing and feature engineering — across **4+** datasets; improved data accuracy by **15%**.
- Validated data quality and consistency, ensuring reliable inputs for downstream analysis and modeling.
- Created statistical predictive models that informed data-driven business decision-making.
- Crafted **Matplotlib** and **Excel** visualizations to communicate insights; cut manual reporting time by **20%**.

Technical Skills

Programming Languages: Python, SQL

Data Science & ML: Power BI, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, SARIMA, Prophet, PyTorch, LSTM, Time Series Forecasting, EDA, Predictive Analytics

AI & Generative AI: LLMs, RAG, AI Agents, Prompt Engineering, Groq API, Llama 3

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Docker, MLflow, VS Code, Jupyter Notebook

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

Projects

Retail Demand Forecasting & Inventory Optimization | *Python, XGBoost, Prophet, PyTorch, FastAPI, Streamlit, Docker* | [URL](#)

- Engineered a forecasting engine on **3.5 years** of retail data across **25** store-department segments; benchmarked **4** architectures (SARIMA, Prophet, LSTM, XGBoost).
- Achieved **4.74% MAPE** using global **XGBoost** with lagged/rolling features, outperforming Prophet (**4.93%**) and LSTM (**10.37%**).
- Optimized EOQ, Safety Stock, and ROP parameters; achieved **15–20% cost savings** in a **26-week** simulation.
- Deployed **2 microservices (FastAPI + Streamlit)** via **Docker Compose**; tracked experiments with **MLflow**.

CarpoolPal – Ride Sharing Application | [URL](#)

- Built a full-stack carpooling platform using **FastAPI, MongoDB**, and **React.js** with **8+** REST endpoints.
- Trained a **Random Forest** model (**100** estimators) on **1,000+** ride records, achieving fare predictions accurate to **12/km**.
- Applied the **Haversine Formula** and secured auth with **JWT/bcrypt**; improved peak-hour fare accuracy by **18%**.

MedAgent — AI Medical Assistant | *Python, Groq API, Llama 3, RAG, Streamlit* | [URL](#)

- Architected an agentic AI medical assistant using **RAG** and the **Llama 3** LLM to answer health queries from medical documents.
- Optimized inference using **Groq's LPU-based** architecture; reduced response time to under **2 seconds** per query, a **4x** speedup.
- Enabled real-time **RAG-based retrieval** across **100+** medical topics, improving query coverage and accuracy.
- Maintained multi-turn conversational context using session memory across **10+** exchanges.

Certificates

- Artificial Intelligence Fundamentals** – IBM SkillsBuild (**2024**)
- Applied Machine Learning in Python** – Coursera (**2023**)
- HackerRank 5-Star** in Python – Solved **100+** algorithmic problems

Leadership / Achievements

- Volunteered with the **NSS Club**, leading **5+** awareness drives and coordinating **3+** campus events.
- Kanchan Education and Welfare Society:** Supported welfare programs reaching **50+** beneficiaries.